

Multiple Choice (10 marks)

1. A student hypothesizes that the amount of water a plant receives will affect how long it grows before reaching maturity. To test this hypothesis, the student...
 - (a) several digital scales.
 - (b) a wet paper towel.
 - (c) several identical plant seeds.
 - (d) different sizes of containers.
2. Find the opposite of the number -7.
 - (a) 7
 - (b) -7
 - (c) 0.7
 - (d) -7.7
3. In MSM, early syphilis was found to be associated with:
 - (a) HIV infection
 - (b) Use of methamphetamine
 - (c) Sex with recent internet partners
 - (d) All of the above
4. An appliance manufacturer has redesigned its original electrical fan. It has made the new fan more energy efficient. This means that the new fan
 - (a) converts more electricity into heat.
 - (b) spins at a slower rate than the original fan.
 - (c) requires more electricity than the original fan.
 - (d) reduces the percentage of heat lost to the atmosphere.
5. Which of the following processes is responsible for changing liquid water into water vapor?
 - (a) photosynthesis
 - (b) condensation
 - (c) evaporation
 - (d) precipitation

True / False (10 marks)

Answer True or False and justify briefly.

1. True or False: The correct answer to 'Which of these items contains a fat soluble vitamin?' is 'dog'.
2. True or False: The correct answer to 'Under Treasury Circular 230, in which of the following situations is a CPA prohibited from giving written advice concerning one or more...' is 'The CPA takes into account the possibility that a tax return will not be audited.'.
3. True or False: The correct answer to 'Which method can be used to solve 11×13 ?' is 'Multiply 11×10 and 11×3 , then add the two products.'.
4. True or False: The correct answer to 'Free radicals are produced by' is 'Normal metabolism'.
5. True or False: The correct answer to 'Fungi' is 'occupy the top of the food chain'.

Long Form Response (20 marks)

Provide thorough reasoning and reference key steps.

1. Write a python function to reverse only the vowels of a given string.
2. Write a python function to check whether the count of divisors is even or odd.

End of Paper